Customer Engagement & Product Utilization Analytics for Retention Strategy

European Bank

Data Analyst Internship Project

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Tools Used:

- Python
- Streamlit
- Pandas
- Plotly
- Microsoft Excel

Objective:
To evaluate how customer engagement, product utilization, and relationship strength influence customer retention and churn behavior.

Submission Year:
2026

1. Introduction

Customer retention is one of the most important drivers of long-term profitability in the banking sector. While financial indicators such as salary and account balance provide useful information, customer engagement and product adoption often play a greater role in determining loyalty.

This project evaluates customer retention through behavioral and relationship-based metrics. The analysis focuses on engagement levels, product utilization, financial commitment, and relationship strength to identify factors associated with customer churn.

The findings support the development of retention strategies, customer engagement programs, and product optimization initiatives.

2. Dataset Overview

The dataset contains customer-level banking information including demographic, financial, and engagement-related attributes.

Key variables analyzed include:

- CustomerId
- Geography
- Gender
- CreditScore
- Age
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Exited

The Exited variable represents customer churn and serves as the primary target variable for retention analysis.

3. Methodology

Data Validation

- Verified customer records
- Validated churn indicators
- Checked product utilization fields
- Reviewed engagement variables

Engagement Analysis

- Compared churn rates of active and inactive customers
- Calculated Engagement Retention Ratio
- Assessed customer activity patterns

Product Utilization Analysis

- Evaluated churn by number of products
- Compared single-product and multi-product customers
- Calculated Product Depth Index

Financial Commitment Analysis

- Evaluated salary versus balance relationships
- Identified high-value disengaged customers
- Detected premium customers at risk

Relationship Strength Assessment

- Developed Relationship Strength Index
- Evaluated impact of credit card ownership
- Identified sticky customer profiles

Executive Insights Development

- Converted analytical findings into actionable retention recommendations

4. Results and Findings

Engagement Analysis

The analysis revealed a significant relationship between customer engagement and churn behavior. Active customers demonstrated substantially lower churn rates than inactive customers, indicating that engagement is a critical predictor of retention.

Product Utilization Analysis

Customers with multiple banking products showed stronger retention compared to customers with only one product. This suggests that deeper product adoption strengthens customer relationships and increases switching costs.

High-Value Customer Analysis

A segment of customers maintained high account balances while remaining inactive. These customers represent an important retention risk because they possess significant financial value but exhibit weak engagement behavior.

Financial Commitment Analysis

Salary and balance analysis demonstrated that financial strength alone does not guarantee retention. Several financially valuable customers exhibited churn risk due to low engagement levels.

Relationship Strength Assessment

Customers with strong engagement, multiple products, and credit card ownership achieved higher Relationship Strength Index scores. These customers demonstrated greater loyalty and retention stability.

Customer Loyalty Analysis

The identification of sticky customer profiles highlighted characteristics associated with long-term retention. Product utilization and engagement consistently emerged as stronger retention indicators than balance alone.

5. Key Insights

1. Active customers exhibit significantly lower churn rates.
2. Product adoption improves customer retention.
3. Engagement is a stronger predictor of loyalty than account balance.
4. High-balance inactive customers represent hidden retention risk.
5. Credit card ownership positively influences customer stickiness.
6. Multi-product customers demonstrate stronger relationship depth.

7. Relationship strength is closely associated with retention performance.
8. Retention strategies should prioritize engagement-driven interventions.

6. Recommendations

Increase Customer Engagement

Develop personalized engagement campaigns targeting inactive customers through digital banking, notifications, and loyalty initiatives.

Expand Product Adoption

Promote cross-selling opportunities to encourage customers to adopt multiple banking products and services.

Retain High-Value Customers

Implement proactive retention programs for high-balance customers exhibiting low engagement levels.

Strengthen Credit Card Programs

Enhance credit card benefits and loyalty rewards to reinforce customer relationships.

Implement Relationship Strength Monitoring

Continuously track customer engagement, product utilization, and relationship strength indicators to identify retention risks early.

Build Predictive Retention Frameworks

Use engagement and product utilization metrics as key inputs for future churn prediction models.

7. Conclusion

The project successfully demonstrated that customer engagement and product utilization play a central role in retention outcomes.

By analyzing engagement behavior, product adoption, financial commitment, and relationship strength, the organization gains actionable insights into customer loyalty and churn risk.

The Streamlit dashboard provides an interactive decision-support tool that enables stakeholders to monitor retention performance and design more effective customer engagement strategies.

8. Executive Summary

This project analyzed customer retention using engagement, product utilization, and relationship-strength indicators.

The analysis demonstrated that active customers and multi-product customers exhibit stronger retention performance. High-balance inactive customers were identified as a critical risk segment requiring targeted retention strategies.

Relationship strength and engagement consistently outperformed financial indicators as predictors of loyalty. These findings support the development of engagement-focused retention programs, product bundling strategies, and proactive customer relationship management initiatives.

The resulting dashboard provides management with a scalable framework for monitoring customer behavior, evaluating churn risk, and improving long-term customer retention.